

Unit 3 & 4 practice questions

1. List postulates of electronic theory of valency
2. State the octet rule
3. List 2 examples of exception to octet rule with less than 8 electrons
4. List 2 examples of exception to octet rule with more than 8 electrons
5. Explain ionic bond with an example.
6. Explain covalent bond with an example
7. Explain coordinate covalent bond with an example
8. Illustrate Lewis dot structure of CH_4 , NH_3 , CO_2 , H_2O , H_2CO_3 , CO_3^{2-}
9. All polar molecules contain polar bonds but all molecules with polar bonds are not necessarily polar- Justify the statement with relevant examples
10. List the criterion of electronegativity values to determine bond types
11. Identify which of the following molecules are ionic, polar or non-polar covalent
 - a) CH_4 (Electronegativity difference 0.4)
 - b) HCl (Electronegativity difference 0.9)
 - c) CO_2 (Electronegativity difference 1.0)
 - d) NaCl (Electronegativity difference 2.23)
12. Define bond strength or bond dissociation energy. Between C-C, C=C and $\text{C}\equiv\text{C}$ which is the strongest.
13. Explain resonance structures with an example.
14. Define colligative property.
15. Explain lowering of vapour pressure along with mathematical expression.
16. Explain elevation of boiling point along with mathematical expression.
17. Explain freezing point depression along with mathematical expression.
18. Define osmosis and reverse osmosis.
19. Explain osmotic pressure along with mathematical expression.
20. All types of numericals done in Unit 4 ppt